Daws86s-Interview Preparation Series(DevSecOps Linux-Day3)

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What is a package manager?

A package manager is a tool that helps install, update, remove, and manage software on a Linux system along with all required dependencies.
Instead of manually downloading binaries, it automates dependency resolution and version management.

Why it matters in DevOps:

- Consistent environments
- Faster automation
- Secure updates

```
apt install nginx
yum install nginx
dnf install nginx
```

Difference between yum and apt

- apt is used in Debian/Ubuntu-based systems
- yum (now replaced by dnf) is used in RHEL/CentOS systems
- Both resolve dependencies automatically
- Difference is package format and ecosystem, not concept

How do you find a file in Linux?

1. If I know the file name → find
2. If I know file content → grep
3. If system index is available → locate

```
find / -name nginx.conf
grep -R "server_name" /etc/nginx
locate nginx.conf
```

What is grep and why is it important?

Answer:

grep is used to search text patterns inside files.
It's critical for log analysis, debugging, and troubleshooting in production.

Real-world use case:

- Finding errors in logs
- Searching config values
- Debugging crashes

```
grep "ERROR" app.log
grep -R "timeout" /var/log
grep -i "failed" logfile
```

What is SSH and how does it work?

Step-by-step explanation:

1. SSH establishes an encrypted connection
2. Client authenticates using password or key
3. Server validates user identity
4. Secure shell session is created

Why DevOps uses SSH:

- Secure remote access
- Automation
- File transfers

```
ssh user@server
ssh -i key.pem user@server
```

How does SSH key-based authentication work internally?

Step-by-step:

1. Private key stays with client
2. Public key is stored on server
3. Server challenges client
4. Client proves key ownership
5. Access granted

Key files:

```
~/.ssh/authorized_keys
```

How do you check Linux logs?

Step-by-step:

1. Check system logs
2. Check service-specific logs
3. Use journalctl for systemd services

Commands:

```
ls /var/log
```

What is systemd and how is it different from init?

Step-by-step:

- init was sequential and slow
- systemd is parallel and dependency-based
- Handles services, logging, timers

Commands:

```
systemctl status
systemctl start nginx
```

Explain Linux process states

Step-by-step explanation:

- Running → executing on CPU
- Sleeping → waiting for IO
- Stopped → paused manually
- Zombie → finished but not cleaned

Command:

```
ps aux
```

How does Linux scheduling work?

Linux uses a scheduler to decide **which process gets CPU time**.

1. All runnable processes are kept in a run queue
2. Scheduler picks processes fairly
3. CPU time is shared based on priority and fairness
4. Context switching happens between processes

How does Linux memory management work?

1. Each process gets virtual memory
2. Virtual memory maps to physical RAM
3. Unused memory is cached
4. Swap is used under pressure

Key idea:
Free memory is wasted memory in Linux.

What is swap and when is it used?

Answer:

Swap is disk space used as **extension of RAM**.

Used when:

- RAM is full
- System needs breathing space

Command:

```
swapon –show
free -m
```

What happens when Linux runs out of memory?

Step-by-step:

1. Linux tries to reclaim cache
2. Uses swap
3. If still exhausted → OOM Killer activates

What is load average?

Answer:

Load average shows:

- Number of processes waiting for CPU or IO

It's **not CPU usage**.

Load average vs CPU utilization

Step-by-step:

- CPU usage → how busy CPU is
- Load average → how many are waiting

High load + low CPU = IO problem

What happens when inodes are exhausted?

Step-by-step:

- Disk shows free space
- New files cannot be created
- Common with many small files

Command:

```
df -i
```

What is /proc filesystem?

Answer:

- Virtual filesystem
- Exposes kernel & process info
- No real files on disk

Command:

```
cat /proc/cpuinfo
cat /proc/meminfo
```

How does cron work internally?

Step-by-step:

1. cron daemon runs in background
2. Reads crontab files
3. Executes jobs at scheduled time
4. Uses minimal environment

Important: cron does NOT load your shell environment.

How do you troubleshoot a Linux server that suddenly becomes slow?

Golden order:

1. Disk
2. Memory
3. CPU
4. Network
5. Application

Commands:

```
df -h
free -m
top
```

How do you check running processes?

Commands:

```
ps aux
top
htop
```

How do you schedule a cron job?

Step-by-step:

1. Write script
2. Give execute permission
3. Add to crontab
4. Redirect logs

Example:

```
crontab -e
* * * * * /script.sh >> /var/log/script.log 2>&1
```

Application running slow – what do you check first?

Answer:

Disk first.
Disk issues cause silent failures and cascading problems.

Deployment failed – how do you debug?

Step-by-step:

1. Check logs
2. Check permissions
3. Check environment variables
4. Check dependency availability

How do you roll back a deployment?

Step-by-step:

1. Restore previous version
2. Verify configs
3. Restart safely
4. Monitor

How do you monitor application logs?

Commands:

```
tail -f app.log
grep ERROR app.log
```

Pipeline fails only on runner, works locally — why?

Step-by-step:

1. Environment mismatch
2. PATH differences
3. Permission issues
4. Missing tools

Script works manually but fails via cron — why?

Reason:

Cron has minimal PATH and environment.

Fix:

- Use absolute paths
- Export variables

Jenkins/GitLab runner offline — Linux checks?

Step-by-step:

1. Disk full
2. CPU exhausted
3. Network issues
4. Runner service status

Docker build fails due to permissions — why?

Answer:

- Files owned by root

Git clone slow on CI but fast locally — why?

Step-by-step:

- Network bandwidth
- DNS latency
- Disk IO

Pipeline hangs indefinitely — Linux checks?

Step-by-step:

1. Check CPU / memory
2. Check stuck process
3. Check disk IO

Runner disk fills up — how to prevent?

Step-by-step:

- Cleanup workspace
- Rotate logs
- Monitor disk

High load only during pipeline execution — why?

Answer:

- Parallel jobs
- Heavy builds
- IO contention

How do you limit CPU/memory for CI jobs at OS level?

```
systemctl set-property service CPUQuota=50%
```

Container exits immediately — Linux-level reasons?

Step-by-step:

1. Main process finished or crashed
2. Missing command / entrypoint
3. Permission issue
4. OOM kill

```
docker logs container
dmesg | grep -i oom
```

Container shows "Killed" — what happened?

Answer:

Usually the **OOM killer** terminated the container due to memory limit breach.

Confirm using:

```
dmesg | grep -i oom
```

Container works on one node but not another — OS checks?

Step-by-step:

1. Kernel version mismatch
2. Filesystem differences
3. Disk pressure
4. Missing kernel modules

How do namespaces isolate processes?

Step-by-step:

- PID namespace → process isolation
- NET namespace → network isolation
- MNT namespace → filesystem isolation

> "Containers are just isolated Linux processes."
>
> ## File permission issues inside container — root cause?
>
> **Step-by-step:**
>
> - UID/GID mismatch
> - Volume mounted from host
> - Files owned by root on host

Why container cannot bind to port <1024?

Answer:

Ports below 1024 require **root privileges**.

Fix:

- Run as root
- Use higher port
- Use capabilities

Disk full inside container but host has space — why?

Step-by-step:

- Container filesystem overlay is full
- Docker storage driver limit
- Logs filling container layer

Kubernetes node Ready but pods not scheduling — Linux checks?

Step-by-step:

1. Disk pressure
2. Memory pressure
3. CPU exhaustion
4. Kubelet logs

Node becomes NotReady — Linux-level investigation

Step-by-step:

1. Check kubelet status
2. Check disk space
3. Check network
4. Check kernel logs

kubelet running but pods failing — why?

Step-by-step:

- CNI issues
- Disk full
- Image pull failures
- Permission issues

DiskPressure / MemoryPressure — Linux metrics to check

Commands:

```
df -h
free -m
df -i
```

Node rebooted — what Linux logs do you check first?

Step-by-step:

1. uptime
2. journalctl
3. dmesg
4. kubelet logs

Pods evicted due to memory — OS explanation

Answer:

- Node under memory pressure
- Kubelet evicts pods to protect node
- Similar to OOM logic

Debugging slow pods caused by node issues

Step-by-step:

1. Check node disk IO
2. Check CPU throttling
3. Check memory pressure

How swap affects Kubernetes (and why disabled)?

Answer:

- Swap hides memory pressure
- Causes unpredictable pod eviction
- Kubernetes prefers fail-fast

EC2 running but SSH not accessible — Linux + cloud checks

Step-by-step:

1. Security group
2. Disk full
3. SSH service
4. Network interface

Disk attached but not visible — Linux steps

Step-by-step:

1. lsblk
2. rescan disks
3. Create filesystem
4. Mount

Instance unreachable after reboot — Linux cause?

Step-by-step:

- fstab misconfiguration
- Filesystem corruption
- Network service failure

CPU credits exhausted — Linux symptoms

Answer:

- High load
- Slow response
- CPU throttling

User-data script partially executed — why?

Step-by-step:

- Script failed midway
- No error handling
- Missing package or network dependency

Best practice:

```
set -e
set -x
```

Cloud VM slow only at night — Linux investigation

Step-by-step:

1. Check cron jobs
2. Check backups
3. Check log rotation
4. Check batch jobs

Network works internally but not externally — OS-level reason

Step-by-step:

- Firewall rules
- NAT configuration
- IP forwarding disabled

Commands:

```
iptables -L
```

node_exporter shows wrong metrics — Linux reason

Step-by-step:

- Permission denied
- Kernel metrics inaccessible
- Old kernel version

Memory looks free but OOM kills happen — why?

Step-by-step:

- Cached memory reclaimed slowly
- cgroup memory limits
- Kernel overcommit

High load average but idle CPU — explanation

Step-by-step:

- Processes blocked on IO
- Disk or network bottleneck

Application slow only during log rotation — why?

Step-by-step:

- Log rotation triggers heavy IO
- Compression causes CPU spike

Monitoring agent consuming high CPU — Linux debugging

Step-by-step:

1. Identify agent process
2. Check scrape frequency
3. Tune collection interval

When do you disable swap and why?

Answer:

- Latency-sensitive systems
- Containers / Kubernetes

Swap hides memory pressure.

Root password lost — recovery steps?

Step-by-step (safe approach):

1. Reboot system
2. Edit GRUB entry
3. Boot into single-user / emergency mode
4. Remount root filesystem as read-write
5. Reset password
6. Reboot normally

Commands:

```
mount -o remount,rw /
passwd
reboot
```

Server stuck in emergency mode — what do you do?

Step-by-step:

1. Read error message shown
2. Check failed mounts (usually fstab issue)
3. Comment incorrect fstab entry
4. Reload system

Hardening checklist for production Linux?

- SSH keys only
- Disable root login
- Firewall enabled
- Regular patching
- Monitoring & alerts

How do you rotate SSH keys safely?

Step-by-step:

1. Add new key
2. Verify access
3. Remove old key
4. Audit access

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